Rajesh Perumal

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Lead software engineer | Agentic AI Platforms

Austin, TX

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Summary

Lead software engineer with 18+ years of experience building high-throughput distributed systems and accelerated enterprise AI applications. Experienced in setting technical direction, mentoring engineers, and eliminating performance bottlenecks to improve system reliability. At AWS Marketplace, led delivery of a unified purchasing workflow and architected a parallelized Graph RAG platform that processes seller contracts with high-accuracy context grounding and citation tracing.

Professional Experience

Software Development Engineer II Jun 2022 - Present
AWS, Amazon Inc. | Austin, TX

- Architected an enterprise-grade Agentic RAG engine utilizing a parallelized, dual-model hybrid architecture; deployed low-cost vector transformations to route intent and targeted Claude Sonnet execution to accelerate end-to-end inference speeds by up to 40% and reduce time-to-first-token (TTFT) when extracting specific clauses from End User License Agreement (EULA).
- Eliminated token context bloat and structural algorithmic bottlenecks in deep learning payloads by engineering a parallel Neo4j and Stardog semantic graph retrieval layer, reducing multi-hop graph traversal latency and lowering context overhead by 35%.
- Architected a parallelized 7-stage data ingestion workflow utilizing Strands Agents SDK (ReAct framework) to orchestrate multi-agent interactions, automated reasoning, and knowledge graph injection into high-throughput graph database (AWS Neptune).
- Executed a zero-downtime, low-level translation of core procurement infrastructure into an IPv6 dual-stack domain within 2 weeks; performed rigorous traffic benchmarking and concurrency load-testing to guarantee zero line-rate throughput degradation for active legacy IPv4 clients.

Senior Software Engineer Aug 2021 - Jun 2022
The Home Depot Inc. | Austin, TX

- Led a platform modernization initiative, establishing CI/CD pipeline standards and reusable HTTP client libraries to improve consistency, maintainability, and service-integration patterns across engineering teams.

Principal Software Engineer Jun 2016 - Aug 2021
Dell Technologies | Round Rock, TX

- Developed full-stack product catalog definitions for configurable items with ASP.NET Core, Angular, Bootstrap, and REST APIs, enabling configurable product experiences.
- Led the migration of monolithic applications to cloud-native microservices on Pivotal Cloud Foundry, integrating frontend applications, backend APIs to deliver scalable customer-facing experiences.

Senior .Net Developer Jun 2014 - May 2016
Valence Health Inc. | Austin, TX

- Built claims-processing and enrollment applications using ASP.NET MVC 4 and REST APIs for healthcare operations workflows.

Lead Software Engineer Nov 2008 - May 2014
Humana Inc. (via TCS) | Louisville, KY

- Built single-page applications with role-based security and WCF SOAP services supporting enterprise healthcare workflows.

Education

Anna University, Chennai, India 2004 - 2008
Bachelor of Technology Information Technology

Core Competencies

Full-Stack & Platform Engineering
React, TypeScript, JavaScript, Next.js, XState, Redux, HTML5/CSS3, Angular, Java, Python, Node.js, C#, .NET Core, REST APIs, GraphQL, Distributed Systems, Event-Driven Architecture, Workflow Orchestration, Go (Golang), Microservices, Design Patterns, Architecture Reviews, Multi-core Hardware Fundamentals

AWS, Cloud & Reliability
AWS, Lambda, Step Functions, EventBridge, S3, DynamoDB, OpenSearch, Fargate, Docker, Kubernetes, IAM, KMS, CloudWatch, Observability, Metrics, Alarms, Performance and Reliability Engineering, AWS CDK, SQL Server, PostgreSQL

AI-Enabled Platform Capabilities
Amazon Bedrock AgentCore, Strands Agents SDK (ReAct Pattern), Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Human-in-the-Loop Workflows, Tool Calling, Citation Grounding, Guardrails, OpenSearch Hybrid Retrieval, Titan Embeddings, Claude Code, Kiro, multi-agent orchestration, Spec driven development, Parallel Programming, High-Throughput Pipelining, Memory Optimization, Large Language Models, Natural Language Processing, Inference Latency Optimization, Context Payload Pruning

Data, Search & Knowledge Systems
Neo4j, Stardog, Cypher, SPARQL, OWL 2, Kafka, PySpark, Apache Iceberg, Property Graphs, Semantic Modeling, Vector Search

Quality & Delivery
Automated Testing, pytest, Jest, React Testing Library, Git, GitLab, Jenkins, TeamCity, Feature Flags, A/B Testing, Secure Authorization, Cross-Functional Delivery, Product and Security Partnership, CI/CD, Staged Rollouts, API Design and Standards, Integration Testing, Production Support, SAST, IaC security, Policy validation, CI/CD security gates, LLM-assisted secure code review, DAST, TDD